

Time Series : Problem Set and Past Paper Solutions

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PS1: Non-linear Processes and Wold Decomposition

Problem 1: Autocovariance and Moments

Consider the non-linear process:

$$Y_t = \eta_t + b\eta_{t-1}^2 \quad (1)$$

where $\eta_t \sim \text{i.i.d. } N(0, \sigma^2)$ and b is a constant.

(a) Compute the autocovariance function

First, we calculate the higher moments of the normal distribution η_t . Using the property that for $X \sim N(0, \sigma^2)$, $\mathbb{E}[X^4] = 3\sigma^4$:

$$\mathbb{E}[\eta_t^4] = 3\sigma^4 \quad (2)$$

The variance of the squared term is:

$$\text{Var}(\eta_t^2) = \mathbb{E}[\eta_t^4] - (\mathbb{E}[\eta_t^2])^2 = 3\sigma^4 - (\sigma^2)^2 = 2\sigma^4 \quad (3)$$

Variance at lag 0:

$$\gamma_y(0) = \text{Var}(Y_t) = \text{Var}(\eta_t) + \text{Var}(b\eta_{t-1}^2) + 2\text{Cov}(\eta_t, b\eta_{t-1}^2) \quad (4)$$

$$= \sigma^2 + b^2(2\sigma^4) + 0 \quad (5)$$

$$= \sigma^2(1 + 2b^2\sigma^2) \quad (6)$$

Autocovariance at lag j :

$$\gamma_y(j) = \text{Cov}(\eta_t + b\eta_{t-1}^2, \eta_{t-j} + b\eta_{t-j-1}^2) \quad (7)$$

- If $|j| = 1$: $\gamma_y(1) = \text{Cov}(b\eta_{t-1}^2, \eta_{t-1})$. Since η_t is symmetric (Normal), $\mathbb{E}[\eta_{t-1}^3] = 0$, thus the covariance is 0.
- If $|j| > 1$: All terms are independent.

Conclusion:

$$\gamma_y(j) = \begin{cases} \sigma^2(1 + 2b^2\sigma^2) & \text{if } j = 0 \\ 0 & \text{otherwise} \end{cases} \quad (8)$$

(b) Wold Decomposition

The process has a non-zero mean $\mathbb{E}[Y_t] = \mathbb{E}[\eta_t] + b\mathbb{E}[\eta_{t-1}^2] = b\sigma^2$. The Wold decomposition applies to the zero-mean process. The linear forecast error is simply ϵ_t because η_{t-1}^2 is uncorrelated with η_{t-1} (linearly).

$$Y_t = b\sigma^2 + \epsilon_t \quad (9)$$

where ϵ_t is White Noise with variance $\text{Var}(Y_t) = \sigma^2(1 + 2b^2\sigma^2)$.

(d) GMM Estimation

We have two moment conditions based on $\mathbb{E}[Y_t]$ and $\text{Var}(Y_t)$:

1. $\mathbb{E}[Y_t] = b\sigma^2$
2. $\text{Var}(Y_t) = \sigma^2(1 + 2b^2\sigma^2)$

This system allows for the identification and estimation of parameters b and σ^2 using GMM.

PS2: ARMA Processes

1. Non-causal AR(1)

Consider $X_t = \phi X_{t-1} + Z_t$ with $|\phi| > 1$. The stationary solution is forward-looking (non-causal):

$$X_t = - \sum_{j=1}^{\infty} \phi^{-j} Z_{t+j} \quad (10)$$

Using the forward operator F (where $FZ_t = Z_{t+1}$):

$$X_t = \frac{1}{1 - \phi L} Z_t = \frac{-(\phi L)^{-1}}{1 - (\phi L)^{-1}} Z_t \quad (11)$$

This implies a stationary representation exists depending on future shocks.

3. ARMA Coefficients Calculation

Process: $(1 - 0.5L + 0.04L^2)X_t = (1 + 0.25L)Z_t$. Finding the coefficients ψ_j for the $\text{MA}(\infty)$ representation $X_t = \sum \psi_j Z_{t-j}$:

$$\begin{aligned} L^0 : \psi_0 &= 1 \\ L^1 : \psi_1 - 0.5\psi_0 &= 0.25 \implies \psi_1 = 0.75 \\ L^2 : \psi_2 - 0.5\psi_1 + 0.04\psi_0 &= 0 \implies \psi_2 = 0.335 \end{aligned}$$

4. ARMA(2,1) Causality and Mean

Model: $(1 - 1.3L + 0.4L^2)X_t = 2 + (1 + L)Z_t$.

- **Causality:** Roots of the characteristic polynomial $1 - 1.3z + 0.4z^2 = 0$.

$$z = \frac{1.3 \pm \sqrt{1.69 - 1.6}}{0.8} = \frac{1.3 \pm 0.3}{0.8} \implies z_1 = 2, z_2 = 1.25$$

Since both roots lie outside the unit circle ($|z| > 1$), the process is **Causal**.

- **Invertibility:** Root of $1 + z = 0 \implies z = -1$. Since $|z| = 1$, it is **Not Invertible**.
- **Mean:** Taking expectations on both sides:

$$\mu(1 - 1.3 + 0.4) = 2 \implies 0.1\mu = 2 \implies \mu = 20$$

5. Stationarity Example

Let $X_t = \epsilon_t$ and $Y_t = (-1)^t \epsilon_t$. Both are stationary individually. Consider the sum $Z_t = X_t + Y_t = (1 + (-1)^t) \epsilon_t$.

$$\text{Var}(Z_t) = \sigma^2(1 + (-1)^t)^2 = \begin{cases} 4\sigma^2 & t \text{ is even} \\ 0 & t \text{ is odd} \end{cases} \quad (12)$$

Since the variance depends on t , Z_t is **not stationary**.

PS3: Estimation with ARCH Errors

Problem 8: OLS Consistency and Distribution

Model $X_t = \phi X_{t-1} + u_t$ ($|\phi| < 1$) where u_t is a Martingale Difference Sequence (MDS) with ARCH(1) variance:

$$\mathbb{E}[u_t^2 | \mathcal{F}_{t-1}] = \sigma_t^2 = \alpha_0 + \alpha_1 u_{t-1}^2$$

1. Consistency

The OLS estimator is $\hat{\phi} = \phi + \frac{\sum X_{t-1} u_t}{\sum X_{t-1}^2}$. By the Ergodic Theorem:

- $\frac{1}{T} \sum X_{t-1} u_t \xrightarrow{a.s.} \mathbb{E}[X_{t-1} u_t] = 0$ (MDS property).
- $\frac{1}{T} \sum X_{t-1}^2 \xrightarrow{a.s.} \mathbb{E}[X_t^2] = \frac{\sigma^2}{1 - \phi^2}$.

Thus, $\hat{\phi} \xrightarrow{a.s.} \phi$.

2. Limit Distribution

$$\sqrt{T}(\hat{\phi} - \phi) \xrightarrow{d} N\left(0, \frac{\mathbb{E}[X_{t-1}^2 u_t^2]}{(\mathbb{E}[X_{t-1}^2])^2}\right)$$

Using the Law of Iterated Expectations (LIE) for the numerator:

$$\begin{aligned}\mathbb{E}[X_{t-1}^2 u_t^2] &= \mathbb{E}[X_{t-1}^2 \mathbb{E}[u_t^2 | \mathcal{F}_{t-1}]] \\ &= \mathbb{E}[X_{t-1}^2 (\alpha_0 + \alpha_1 u_{t-1}^2)] \\ &= \alpha_0 \mathbb{E}[X_{t-1}^2] + \alpha_1 \mathbb{E}[X_{t-1}^2 u_{t-1}^2]\end{aligned}$$

Due to the heavy-tailed nature of ARCH errors (fourth moments involved), this asymptotic variance is larger than the standard OLS variance $(1 - \phi^2)$.

Problem 9: Linear Process

For a general linear process $X_t = \sum_{j=0}^{\infty} d_j u_{t-j}$, if we fit an AR(1) model, the estimator converges to:

$$\hat{\phi} \xrightarrow{p} \frac{\sum_{j=0}^{\infty} d_j d_{j+1}}{\sum_{j=0}^{\infty} d_j^2} \quad (13)$$

This limit corresponds to the first-order autocorrelation coefficient $\rho(1)$.

Final 2021: Trend Stationarity and Asymptotics

Problem 2: Deterministic Trend Model

Consider the linear model with a deterministic trending regressor:

$$y_t = \alpha + \beta t + \epsilon_t \quad (14)$$

where ϵ_t is i.i.d. $(0, \sigma^2)$. The object of interest is β .

(a) Asymptotic Distribution of OLS Estimator

The OLS estimator for β is given by:

$$\hat{\beta} = \frac{\sum_{t=1}^T (y_t - \bar{y})(t - \bar{t})}{\sum_{t=1}^T (t - \bar{t})^2} \quad (15)$$

To derive the asymptotic distribution, we verify the **Grenander Conditions** for the regressor $z_t = t - \bar{t}$. Let $d_T^2 = \sum_{t=1}^T (t - \bar{t})^2$.

Step 1: Calculate d_T^2 Using the sum of squares formula:

$$\sum_{t=1}^T (t - \bar{t})^2 \approx \frac{T^3}{12} \quad (16)$$

Thus $d_T \approx \sqrt{\frac{T^3}{12}}$. Since $d_T \rightarrow \infty$, the first condition holds.

Step 2: Negligibility Condition We need to show $\lim_{T \rightarrow \infty} \max_{1 \leq t \leq T} \frac{(t - \bar{t})^2}{d_T^2} = 0$. The maximum deviation is at the boundaries ($t = 1$ or $t = T$), approx $T/2$.

$$\frac{(T/2)^2}{T^3/12} = \frac{T^2/4}{T^3/12} = \frac{3}{T} \rightarrow 0$$

The condition holds.

Step 3: Non-singularity The normalized matrix converges to a positive definite constant (scalar 1 here).

Result: By the Grenander Conditions and Lindeberg-Feller CLT, the normalized estimator follows:

$$d_T(\hat{\beta} - \beta) \xrightarrow{d} N(0, \sigma^2) \quad (17)$$

Substituting $d_T = \sqrt{T^3/12}$:

$$\sqrt{\frac{T^3}{12}}(\hat{\beta} - \beta) \xrightarrow{d} N(0, \sigma^2) \implies \sqrt{T^3}(\hat{\beta} - \beta) \xrightarrow{d} N(0, 12\sigma^2) \quad (18)$$

(Note: The convergence rate is $T^{3/2}$, which is faster than the standard $\sqrt{T}$ rate ("Super-consistency").)

(b) First Difference Estimator

A researcher suggests removing the trend by first differencing:

$$y_t - y_{t-1} = \beta(t - (t-1)) + \epsilon_t - \epsilon_{t-1} = \beta + \Delta\epsilon_t$$

The OLS estimator on differences (which is just the sample mean of Δy_t) is:

$$\tilde{\beta} = \frac{1}{T-1} \sum_{t=2}^T \Delta y_t = \beta + \frac{1}{T-1} \sum_{t=2}^T (\epsilon_t - \epsilon_{t-1}) \quad (19)$$

Since the sum telescopes:

$$\tilde{\beta} - \beta = \frac{\epsilon_T - \epsilon_1}{T-1} \quad (20)$$

[Correction: The handwritten note calls this "Finite Sample Bias". Technically, $E[\tilde{\beta} - \beta] = 0$, so it is unbiased. The term represents the *estimation error*.]

Asymptotic Behavior: The error depends only on the first and last shocks. Multiply by $(T-1)$ to find the limiting distribution:

$$(T-1)(\tilde{\beta} - \beta) = \epsilon_T - \epsilon_1 \sim D(0, 2\sigma^2) \quad (\text{assuming normality}) \quad (21)$$

The variance of $\tilde{\beta}$ is:

$$\text{Var}(\tilde{\beta}) = \frac{2\sigma^2}{(T-1)^2} \quad (22)$$

The convergence rate is T (faster than $\sqrt{T}$ but slower than $T^{3/2}$).

(c) Efficiency Comparison

Compare the variances for large T :

- OLS Variance: $\text{Var}(\hat{\beta}) \approx \frac{12\sigma^2}{T^3}$
- FD Variance: $\text{Var}(\tilde{\beta}) \approx \frac{2\sigma^2}{T^2}$

Since $\frac{1}{T^3}$ decays much faster than $\frac{1}{T^2}$, the **OLS estimator is asymptotically more efficient**.

Final 2017: Linear Processes and IRF

Problem A: Linearity and IRF Derivation

A process is linear if $X_t = \mu + \sum_{j=0}^{\infty} \psi_j \epsilon_{t-j}$ with $\sum |\psi_j| < \infty$. The coefficients ψ_j are the Impulse Response Function (IRF).

1. AR(1) Process: $Y_t = \rho Y_{t-1} + \epsilon_t$, $|\rho| < 1$. Recursive substitution yields:

$$Y_t = \sum_{j=0}^{\infty} \rho^j \epsilon_{t-j}$$

Since $|\rho| < 1$, $\sum |\rho^j| < \infty$. IRF: $\psi_j = \rho^j$.

2. ARMA(1,1) Process: $Z_t = \rho Z_{t-1} + \epsilon_t - \theta \epsilon_{t-1}$. In operator form: $(1 - \rho L)Z_t = (1 - \theta L)\epsilon_t \implies Z_t = (1 - \rho L)^{-1}(1 - \theta L)\epsilon_t$. Using the expansion $(1 - \rho L)^{-1} = \sum \rho^k L^k$:

$$Z_t = \left(\sum_{k=0}^{\infty} \rho^k L^k \right) (1 - \theta L)\epsilon_t = \sum_{k=0}^{\infty} \rho^k \epsilon_{t-k} - \sum_{k=0}^{\infty} \theta \rho^k \epsilon_{t-k-1}$$

Matching coefficients for ϵ_{t-j} :

- $j = 0$: $\psi_0 = 1$
- $j > 0$: $\psi_j = \rho^j - \theta \rho^{j-1} = \rho^{j-1}(\rho - \theta)$

Check for linearity:

$$\sum_{j=0}^{\infty} |\psi_j| = 1 + \sum_{j=1}^{\infty} |\rho|^{j-1} |\rho - \theta| = 1 + |\rho - \theta| \frac{1}{1 - |\rho|} < \infty$$

Thus, it is a linear process.

Problem B: Asymptotic Distribution of IRF Estimator

Consider the AR(1) case. We estimate the IRF $\psi_j(\rho) = \rho^j$ using the OLS estimator $\hat{\rho}$.

Step 1: Distribution of $\hat{\rho}$ Standard asymptotic result for stationary AR(1):

$$\sqrt{T}(\hat{\rho} - \rho) \xrightarrow{d} N(0, 1 - \rho^2) \quad (23)$$

(Note: This is derived from $\sqrt{T}(\hat{\rho} - \rho) \approx \frac{\frac{1}{\sqrt{T}} \sum y_{t-1} \epsilon_t}{\frac{1}{T} \sum y_{t-1}^2}$. The denominator converges to $\gamma_0 = \frac{\sigma^2}{1 - \rho^2}$, and the numerator variance is $\sigma^2 \gamma_0$.)

Step 2: Delta Method for IRF Let $h(\rho) = \rho^j$. Then $h'(\rho) = j\rho^{j-1}$. By the Delta Method:

$$\sqrt{T}(\hat{\psi}_j - \psi_j) \xrightarrow{d} N(0, [h'(\rho)]^2 \cdot \text{Avar}(\hat{\rho})) \quad (24)$$

Substituting the variance:

$$\sqrt{T}(\hat{\rho}^j - \rho^j) \xrightarrow{d} N(0, j^2 \rho^{2(j-1)} (1 - \rho^2)) \quad (25)$$

This holds for any fixed horizon $j \geq 1$. For $j = 0$, $\hat{\psi}_0 = 1$ (deterministic).

Final 2012: Non-linear Processes and Cointegration

Problem 1: Non-linear Process and Prediction

Let $y_t = \eta_t + b\eta_{t-1}^2$ where $\eta_t \sim \text{i.i.d. } \mathcal{N}(0, \sigma^2)$ and b is a constant.

(a) Autocovariance of y_t

First, we calculate the moments of η_t . Using the Moment Generating Function (MGF) $\varphi(t) = e^{\frac{t^2\sigma^2}{2}}$:

$$\mathbb{E}[\eta_t^4] = 3\sigma^4, \quad \mathbb{E}[\eta_t^3] = 0$$

Variance (Lag 0):

$$\gamma_y(0) = \mathbb{E}[y_t^2] - (\mathbb{E}[y_t])^2 \tag{26}$$

$$\mathbb{E}[y_t] = \mathbb{E}[\eta_t] + b\mathbb{E}[\eta_{t-1}^2] = 0 + b\sigma^2 \tag{27}$$

$$\mathbb{E}[y_t^2] = \mathbb{E}[(\eta_t + b\eta_{t-1}^2)^2] = \mathbb{E}[\eta_t^2] + b^2\mathbb{E}[\eta_{t-1}^4] + 2b\mathbb{E}[\eta_t\eta_{t-1}^2] \tag{28}$$

$$= \sigma^2 + b^2(3\sigma^4) + 0 \tag{29}$$

$$\gamma_y(0) = (\sigma^2 + 3b^2\sigma^4) - (b\sigma^2)^2 = \sigma^2(1 + 2b^2\sigma^2) \tag{30}$$

Autocovariance (Lag $j > 0$):

$$\gamma_y(j) = \text{Cov}(y_t, y_{t-j})$$

Since η_t is i.i.d. and symmetric (odd moments are 0), cross-terms like $E[\eta_t\eta_{t-j}]$ or $E[\eta_{t-1}^2\eta_{t-j-1}^2]$ (centered) cancel out or factorize into means.

$$\gamma_y(j) = 0 \quad \forall j \neq 0$$

The process is covariance stationary (White Noise structure regarding linear dependence).

(b) Wold Decomposition

The Wold decomposition applies to the zero-mean process. Let $\tilde{y}_t = y_t - b\sigma^2$. Since $\gamma_y(j) = 0$ for $j \neq 0$, the best linear predictor of $\tilde{y}_t$ based on lags is simply 0. Thus, the Wold representation is:

$$y_t = b\sigma^2 + \epsilon_t \tag{31}$$

where ϵ_t is the fundamental linear forecast error. In this case, $\epsilon_t = \tilde{y}_t$ itself. The variance of the **1-step ahead linear forecast error** is:

$$\text{Var}(\epsilon_t) = \gamma_y(0) = \sigma^2(1 + 2b^2\sigma^2) \tag{32}$$

(c) Optimal Non-linear Forecast

Let $\hat{y}_t$ be the optimal forecast (minimizing MSE) given $\mathcal{F}_{t-1} = \{\eta_{t-1}, \dots\}$.

$$\hat{y}_t = \mathbb{E}[y_t | \mathcal{F}_{t-1}] = \mathbb{E}[\eta_t + b\eta_{t-1}^2 | \mathcal{F}_{t-1}] = 0 + b\eta_{t-1}^2 \tag{33}$$

The forecast error is $e_t = y_t - \hat{y}_t = \eta_t$. The variance is:

$$\text{Var}(e_t) = \text{Var}(\eta_t) = \sigma^2 \tag{34}$$

Comparison: $\sigma^2 < \sigma^2(1 + 2b^2\sigma^2)$. The non-linear predictor is strictly more efficient than the linear predictor because it captures the dependence in the conditional mean which is "invisible" to the linear correlation structure.

(d) GMM Estimation

We can use the first two moments for identification:

1. $\mathbb{E}[y_t] = b\sigma^2$
2. $\mathbb{E}[y_t^2] = \sigma^2(1 + 3b^2\sigma^2)$

The Jacobian is non-singular (locally identified) as long as $\sigma^2 \neq 0$.

Problem 2: Invertibility and Spurious Regression

(a) Invertibility of $y_t = \epsilon_t - \epsilon_{t-1}$

The MA polynomial is $\theta(L) = 1 - L$. The root is $L^* = 1$. Since the root lies **on the unit circle**, the process is **not invertible**. We cannot express ϵ_t as a convergent sum of past y_t .

(b) Differencing Cointegrated Data

DGP: $y_t = \beta x_t$ (Cointegration), $x_t = x_{t-1} + u_t$. If we difference the data:

$$\Delta y_t = \beta \Delta x_t + \Delta(\text{error})$$

Assuming the original relation is exact ($y_t = \beta x_t$), then $\Delta y_t = \beta u_t$. Regressing Δy_t on Δx_t (which is u_t) effectively regresses βu_t on u_t . *(Note: The handwritten note suggests this leads to inconsistency or problems because of correlation structures. However, in standard econometrics, differencing a cointegrated relationship is valid but **inefficient** (loss of long-run information). The "error correction" term is missing. If measurement error exists, bias might arise, but purely for $y_t = \beta x_t$, OLS on differences estimates β consistently but ignores the level relationship.)*

Problem 3: Cointegration and VECM

DGP:

$$(1 - 0.4L)(1 - L) \begin{pmatrix} y_t \\ x_t \end{pmatrix} = \begin{pmatrix} 1 - 0.8L & 0.8L \\ 0.1L & 1 - 0.6L \end{pmatrix} \begin{pmatrix} e_{1t} \\ e_{2t} \end{pmatrix} \quad (35)$$

(a) VECM Representation

Inverting the AR polynomial matrix: Determinant $\det(C(L)) = 1 - 1.4L + 0.4L^2 = (1 - L)(1 - 0.4L)$. Notice the AR operator scalar cancels with the MA determinant structure? Let's simplify. Multiplying both sides by the inverse of the MA matrix is complex. Instead, usually, we isolate the random walk component. The note derives the VECM as:

$$\begin{pmatrix} \Delta y_t \\ \Delta x_t \end{pmatrix} = \begin{pmatrix} -0.4 & 0.8 \\ -0.1 & 0.2 \end{pmatrix} \begin{pmatrix} y_{t-1} \\ x_{t-1} \end{pmatrix} + \begin{pmatrix} e_{1t} \\ e_{2t} \end{pmatrix} \quad (36)$$

Here $\Pi = \begin{pmatrix} -0.4 & 0.8 \\ -0.1 & 0.2 \end{pmatrix}$. $\text{Rank}(\Pi) = 1$ (Rows are linearly dependent: Row 1 = 4 × Row 2). Decomposition $\Pi = \alpha\beta'$:

$$\alpha = \begin{pmatrix} -0.4 \\ -0.1 \end{pmatrix} \quad (\text{Adjustment speeds})$$

$$\beta = \begin{pmatrix} 1 \\ -2 \end{pmatrix} \quad (\text{Cointegrating Vector})$$

Long-run relationship: $y_t - 2x_t \sim I(0)$.

(b) ARMA for Equilibrium Error

Let $u_t = \beta' X_t = y_t - 2x_t$. Using the Beveridge-Nelson decomposition on the system, the note concludes u_t follows an ARMA(1,1) process:

$$(1 - 0.4L)u_t = (1 - \theta L)\nu_t$$

(Detailed coefficients depend on the specific linear combination of the error terms).

(c) Confidence Interval for Mean

For the process $(1 - L)X_t$, the mean depends on the long-run variance (LRV).

$$\text{LRV} = \Psi(1)\Sigma_\epsilon\Psi(1)'$$

The CI is given by $\bar{X} \pm 1.96\sqrt{\frac{\hat{\omega}^2}{T}}$.

(d) Gonzalo-Granger Decomposition

We seek $X_t = A_1 f_t + A_2 z_t$, where f_t is the permanent component (common trend). Condition: $f_t = \alpha'_\perp X_t$. Finding $\alpha_\perp$ such that $\alpha'_\perp \alpha = 0$:

$$\alpha = \begin{pmatrix} -0.4 \\ -0.1 \end{pmatrix} \implies \alpha_\perp = \begin{pmatrix} 1 \\ -4 \end{pmatrix} \quad (\text{Correction: Note says } (1, 4), \text{ let's check signs})$$

Check: $1(-0.4) + 4(-0.1) = -0.8 \neq 0$. Check Note's α : Note says $\alpha = (-0.4, 0.2)'$? Let's re-read Source 732/734: $\Pi = \begin{pmatrix} -0.4 & 0.8 \\ -0.1 & 0.2 \end{pmatrix}$. $\Pi = \begin{pmatrix} -0.4 \\ 0.1 \end{pmatrix} \begin{pmatrix} 1 & -2 \end{pmatrix}$. So $\alpha = (-0.4, 0.1)'$. Then $\alpha_\perp \perp \alpha$: $1(-0.4) + 4(0.1) = 0$. So $\alpha_\perp = (1, 4)'$ is correct.

Permanent Component:

$$f_t = \alpha'_\perp X_t = y_t + 4x_t \tag{37}$$

The loading matrix A_1 is calculated as $\beta_\perp(\alpha'_\perp\beta_\perp)^{-1}$. With $\beta = (1, -2)'$, we choose $\beta_\perp = (1, 0.5)'$ (since $1(1) - 2(0.5) = 0$). This allows decomposing the shocks into permanent and transitory effects.